Unit 3 Stoichiometry Test - Teacher Version / Answer Key (verC)

Subject: chemistry
Course Code: SCH3U
Unit: Unit 3 Stoichiometry Test

Answer Key

1. [stoichiometry | written_response | 2 marks] 1.20 mol Zn3(PO4)2 approximately
2. [stoichiometry | written_response | 2 marks] 146 g Al2(SO4)3 approximately
3. [stoichiometry | written_response | 3 marks] 1.35 x 10^24 hydrogen atoms approximately
4. [stoichiometry | written_response | 8 marks] Empirical formula C2H6O; molecular formula C4H12O2
5. [stoichiometry | written_response | 4 marks] 32.6 g FeCl3 approximately after using 2Fe + 3Cl2 -> 2FeCl3
6. [stoichiometry | written_response | 10 marks] 2H3PO4 + 3Ca(OH)2 -> Ca3(PO4)2 + 6H2O; 10.3 g Ca3(PO4)2 approximately
7. [stoichiometry | application | 6 marks] AgNO3 + NaCl -> AgCl + NaNO3; actual precipitate mass = 1.34 g; theoretical yield is about 1.37 g AgCl; percentage yield is about 98.0%

Student Paper Reference

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Instructions: Answer all questions. Show full work where required. Write clearly and use appropriate notation.

Foundational Stoichiometry Calculations

Section Plan: 3 questions | 7 marks

Instructions: Show all work. Include units and use appropriate significant digits.

1. Calculate the amount of substance in moles for 7.22 x 10^23 formula units of zinc phosphate, Zn3(PO4)2(s). (2 marks)

2. Calculate the mass of 0.415 mol of aluminum sulfate, Al2(SO4)3(s). (2 marks)

3. How many hydrogen atoms are present in 0.750 mol of ammonia, NH3(g)? (3 marks)

Composition and Reaction Stoichiometry

Section Plan: 3 questions | 22 marks

Instructions: Show a complete stoichiometric setup. Write and balance any required chemical equation before solving.

4. A compound is 52.2% carbon, 13.1% hydrogen, and 34.7% oxygen by mass. Its molar mass is 92.1 g/mol. Determine the molecular formula of the compound. (8 marks)

5. In excess chlorine gas, iron reacts to form iron(III) chloride. Calculate the mass of $FeCl_3(s)$ produced when 11.2 g of iron is used. (4 marks)

6. Phosphoric acid reacts with calcium hydroxide to produce calcium phosphate and water. If 6.50 g of $H_3PO_4(aq)$ reacts completely with excess $Ca(OH)_2(aq)$, what mass of $Ca_3(PO_4)_2(s)$ is produced? Write a balanced chemical equation for the reaction. (10 marks)

Lab Yield Analysis

Section Plan: 1 question | 6 marks

Instructions: Use the data provided to determine theoretical yield and percent yield. Show your setup clearly.

7. Use the lab data below to determine the percentage yield of silver chloride, $AgCl(s)$, produced when excess sodium chloride is added to silver nitrate solution. Lab data: mass of $AgNO_3$ used = 1.62 g; mass of clean, dry filter paper = 1.14 g; mass of dry precipitate and filter paper = 2.48 g. Write the balanced chemical equation for the reaction. (6 marks)